

ECON 308 — International Economic Relations

Quiz 1 | September 21, 2026

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Part A: Multiple Choice (5 questions × 2 points = 10 points)

1. In the expenditure approach to GDP, Canada's national income identity is

$$Y = C + I + G + (X - M).$$

Suppose exports are \$800 billion and imports are \$840 billion, with all other components (C, I, G) held constant. Which statement *best* describes Canada's trade balance and its *direct* effect on GDP?

- A) Trade *surplus* of \$40 billion; this adds \$40 billion to GDP.
 - B) Trade *deficit* of \$40 billion; this reduces GDP by \$40 billion.
 - C) Trade *deficit* of \$40 billion; this reduces GDP by \$80 billion because both exports and imports enter the identity separately.
 - D) Trade *surplus* of \$840 billion; imports create income for trading partners and therefore do not reduce domestic GDP.
2. The simple gravity model predicts bilateral trade as

$$T_{ij} = A \cdot \frac{Y_i \cdot Y_j}{D_{ij}},$$

where $\alpha = \beta = \gamma = 1$ and $A = 1$. Suppose Country i 's GDP increases by 20%, while Country j 's GDP and the bilateral distance D_{ij} remain unchanged. Predicted bilateral trade T_{ij} will:

- A) Increase by exactly 20%.
- B) Increase by 40%, since GDP growth compounds across both the exporting and importing directions.
- C) Increase by approximately 44%, due to a multiplier effect from economic expansion.
- D) Decrease by 20%, as larger economies become more self-sufficient and require fewer imports.

3. McCallum (1995) compared interprovincial trade within Canada to Canada–U.S. trade at the same physical distance, *after* the Canada–U.S. Free Trade Agreement (CUSFTA) had already eliminated most bilateral tariffs. He found that Canadian provinces traded approximately **20 times more** with each other than with equally distant U.S. states. The most accurate economic conclusion from this finding is:
- A) Physical transport costs are the primary remaining barrier to Canada–U.S. trade.
 - B) Canadian interprovincial trade data systematically overstates actual trade volumes due to measurement differences.
 - C) National borders impose substantial non-tariff costs on trade, even under free trade agreements.
 - D) The CUSFTA agreement failed to liberalize Canada–U.S. merchandise trade in any meaningful way.
4. According to Lecture 1, which of the following *best* describes the historical pattern of world trade as a share of global output since 1870?
- A) World trade as a share of global output rose continuously from 1870 to the present, driven by steady improvements in transportation technology.
 - B) World trade expanded from 1870 to 1914, then collapsed between 1914 and 1945 due to war and protectionism, before recovering and expanding again after World War II.
 - C) World trade was negligible before 1945 and only became significant after the invention of containerization in the late 1950s.
 - D) World trade grew rapidly from 1870 to 1914, then recovered immediately after World War I and continued rising without interruption through to the present.
5. The “Dutch Disease” mechanism, as discussed in Lecture 1, describes how a resource-sector boom can harm the manufacturing sector. Which of the following sequences *best* describes this mechanism in the Canadian context (2002–2014)?
- A) A commodity boom diverts capital investment away from manufacturing, directly reducing manufacturing output through factor-market reallocation.
 - B) A commodity boom causes the Bank of Canada to raise interest rates, increasing financing costs and depressing manufacturing investment.
 - C) A commodity boom improves the trade surplus so dramatically that foreign-produced manufactures flood the domestic market, eliminating domestic manufacturing.
 - D) A commodity boom raises foreign demand for the Canadian dollar, causing the CAD to appreciate, which makes Canadian manufactured exports more expensive for foreign buyers and reduces manufacturing competitiveness.

Part B: Numerical Problem (10 points)**The Gravity Model and Canada's Trade Policy**

Canada's GDP is **\$2,400 billion**. Consider Canada's trade relationships with three partners:

Partner	GDP (\$B)	Distance from Canada (km)
Australia	1,600	12,000
Mexico	1,200	3,000
United States	25,000	1,000

Use the following gravity model *throughout* this problem:

$$T_{ij} = A \cdot \frac{Y_i \cdot Y_j}{D_{ij}}, \quad A = 1 \quad (\text{values in \$billions and km})$$

(a) (2 points) Compute the gravity model's predicted bilateral trade for each country pair: T_{CA} (Canada–Australia), T_{CM} (Canada–Mexico), and T_{CU} (Canada–United States).

(b) (2 points) Calculate the ratio T_{CU} / T_{CA} . Decompose this ratio into the two gravity-model factors—*size* and *distance*—that explain why Canada trades so much more with the U.S. than with Australia. Interpret each factor.

(c) (2 points) CPTPP membership reduces Canada's effective distance to Australia by **25%**, from 12,000 km to 9,000 km. Compute the new predicted trade T_{CA}^{CPTPP} and the percentage increase relative to the baseline in part (a).

(d) (4 points) Suppose Canada's initial GDP is $Y = \$2,400$ billion, with net exports $NX = +\$48$ billion. A U.S. recession causes Canadian exports to fall by **\$96 billion**, while imports (M), consumption (C), investment (I), and government spending (G) all remain unchanged. Using the expenditure identity $Y = C + I + G + NX$:

(i) Calculate the new level of net exports after the shock.

(ii) Calculate the new level of Canadian GDP.